

THE MICROPROCESSORS & MICROCONTROLLERS

Instructor: MSc. The Tung Than

PRACTICE EXERCISE #4:

USING UART

I. Student preparation

- Knowledge of how to install and use UART.
- Knowledge of how to use Serial UART LCD

II. Practice content (6 points)

1. Design a 4x4 keyboard set including the following buttons (2 points):
 - From 0 to 9
 - The signs + - * /
 - Sign =
 - Reset button
2. Using AT89C51/AT89C52 in combination with the module just designed above to design a handheld calculator, display calculations and results on an LCD that receives data via UART (4 points).

III. Exercises (4 points)

3. Split the system into two AT89C51 microcontrollers: the Slave manages the 4×4 keypad and transmits key codes via UART, while the Master receives the data, processes the calculations, and sends the result to the LCD display.

Tasks:

- Slave: scan the keypad, encode each key press into 1 byte (0x00–0x0F), and transmit via UART at 9600 bps.
- Master: receive bytes from the Slave via UART, decode them, and process the arithmetic expression.
- Master sends the result string to the UART LCD using a separate UART channel.
- Use debug LEDs: Slave blinks when transmitting, Master blinks when a byte is successfully received.
- Handle connection loss: 2-second timeout — display "COMM ERR" on the LCD if no data is received.

IV. Report

Compress design files and report files into a file named as follows:

[<LAB...>]-[<Student code>]-Full name

The required report file contains the following contents:

1. Design result (screenshot and pasted in the report).
2. Explain the operating principle of the effects, accompanied by a video (send a Google Drive link) to demonstrate the circuit operation in case the instructor cannot run the design file.
3. Exercise report.